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DFT study on gas-phase decomposition of ethylene carbonate in the presence of $LiPF_6$, $LiBF_4$, PF_6^-, and BF_4^-

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Abstract. The decomposition of Li-ion battery (LIB) electrolyte has been a well-known challenge that needs to be overcome. The most common electrolyte on lithium-ion batteries is $LiPF_6$ which has all-balanced properties, while $LiBF_4$ has been proven for its superior stability. These lithium salts are often dissolved in Ethylene Carbonate (EC) to form liquid electrolyte systems. In this work, we investigate the decomposition mechanism of EC in the presence of $LiPF_6$, $LiBF_4$, and their delithiated counterpart by means of first-principles density functional theory (DFT) calculations. We found that the energy barrier of decomposition on $LiBF_4$ presence is 0.42 eV lower than on LiPF6 presence, also on BF_4^- presence is 0.22 eV lower than PF_6^- presence. This suggests that $LiBF_4$ and BF_4^- presence reduces EC stability more than $LiPF_6$ and PF_6^- anion. Moreover, the presence of Li^+ ion increases the energy barrier of decomposition (about 0.79 eV on PF_6^- case, 0.59 eV on BF_4^- case) but decreases enthalpy change significantly (about 1.58 eV on PF_6^- case, 1.43 eV on BF_4^- case). This suggests that while the Li^+ ion causes the decomposition to be slower, its presence destabilizes the EC more.

Keywords: Decomposition mechanism, ethylene carbonate, lithium salt, lithium-ion batteries.

1. Introduction

Lithium-ion batteries (LIB) are widely used as a power source in nowadays life. The importance of LIB as a power source for varieties of electronic devices is due to its high energy density and affordable price at same the time. However, it is also well-known that the energy density stored in a LIB cell deteriorates over time [1]. The deterioration of a LIB cell could be summarized as due to: loss of lithium inventory (LLI), loss of anode/cathode active materials (LAM), internal resistance increase (RI), and loss of electrolyte (LE). From these degradation phenomena, except for the LLI, all of them may be caused by electrolyte decomposition [2]. Hence, electrolyte decomposition has become a problem that attracts much attention due to its direct effect on LIB lifetime.

One of the most common electrolytes used in commercial LIB is Lithium hexafluorophosphate ($LiPF_6$), which has the most balanced properties (i.e. conductivity, thermal stability, anodic stability, etc.) [3,4]. On other hand, a less popular electrolyte, Lithium tetrafluoroborate ($LiBF_4$), has shown better thermal stability [5–8] and improved performance at low temperatures [9] although lower conductivity [7,10]. Furthermore, these lithium salt electrolytes need a highly polar solvent for them to be liquid, where the most widely used solvent is ethylene carbonate (EC) because of its high dissociation ability and ionic conductivity [11]. While $LiPF_6$ and $LiBF_4$ in EC solvent have been frequently studied experimentally, the decomposition mechanisms of these systems are still not well understood.

In this work, we use density functional calculation to study the initial decomposition mechanism of EC in the presence of $LiPF_6$ dan $LiBF_4$ salts. We compare the relative stability of the interaction between EC and these two salts. We also discuss the role of Li ion on the EC decomposition mechanism.

2. Computational details

We modelled the system as one-on-one interaction between molecules in the gas phase. Density functional theory (DFT) calculations were employed using the Gaussian 09 program package [12]. The Beckee Three Hybrid Parameter Exchange - Lee, Yang, and Parr Correlation (B3LYP) method was used as the DFT Exchange-Correlation functional, with chosen basis sets are 6-31G++(d,p). Structure optimizations were performed using the Berny algorithm for finding stable states and the Quadratic Synchronous Transit (QST) approach for finding transition states.

The initial and final states were verified for having real frequency only. On the contrary, the transition states were verified for having an imaginary frequency. This means that the transition states were at saddle points on the potential energy surfaces, indicating the energy barrier of the path between initial and final states. While the energy difference of the final state from the initial state indicates the enthalpy change.

3. Results and discussion

3.1. Decomposition mechanism

The decomposition of EC may have many paths with various final products, but we take the first stage of the decomposition reaction in which its final product is achieved by only 1 transition state. Thus, we first investigate the most stable configuration of EC and lithium salt molecules for determining the possible final state by only 1 transition state. Here, we have 4 cases of initial state: EC + $LiPF_6$, EC + $LiBF_4$, EC + PF_6^-, and EC + BF_4^-. **Figure 1** shows the most stable configurations for every initial state case.

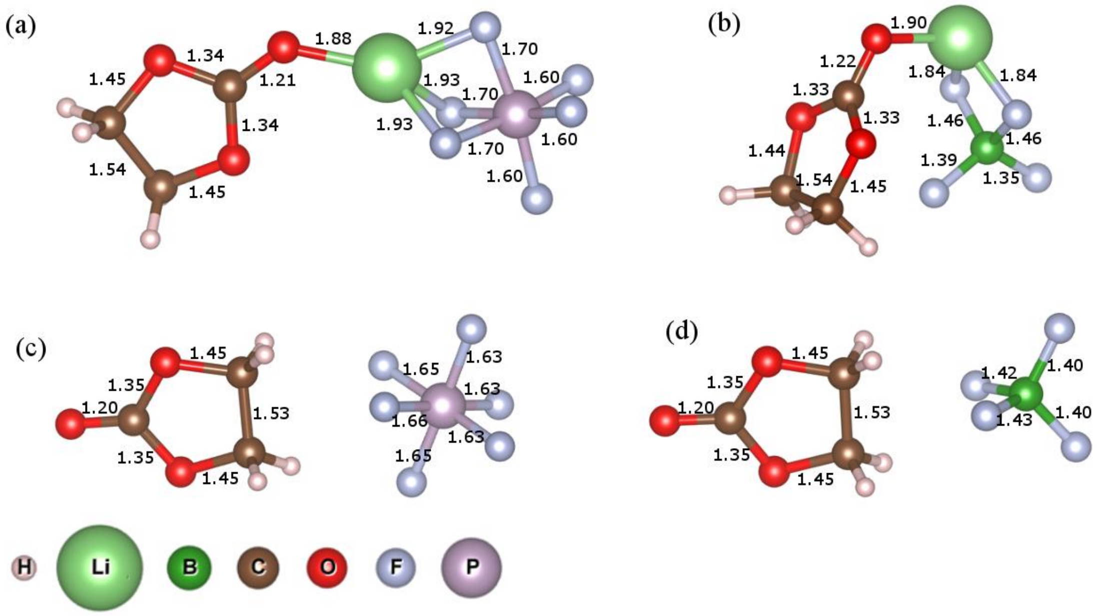

Figure 1. Initial states of EC + (a)$LiPF_6$, (b)$LiBF_4$, (c)PF_6^-, and (d)BF_4^- with bond lengths (Å).

As we can see in **Figure 1**, the lithium-ion attracts strongly (as seen from similar bond lengths of Li-O and Li-F) to the carbonate side of EC while in the delithiated case the ethylene side of EC attracts the lithium salt anion. All of them show a detached lithium salt anion (PF_6^- or BF_4^-) which can become stable by removing one of the Fluor ions and turning it into lewis acid (PF_5 or BF_3). This suggests that the first stage of EC decomposition is the breaking of EC structure by the invasion of F^- ion. By this assumption, we search for stable structures constructed by an F- ion bonds into the EC molecule and break the initial EC structure.

Now the problem is to determine which bond in EC that breaks. Several studies have proposed that decomposing EC will undergo an O_E-C_E (ethylene oxygen and ethylene carbon) bond dissociation [13–17]. Then, we have two possible places for the F^- ion to bonds, which are O_E or C_E. Nevertheless, the former is very unlikely to happen because both O and F atoms have high electronegativity, while the C-F bond is a common product in hydrocarbon substitutions. Consequently, the resulting reaction will yield a 2-fluoroethyl carbonate structure.

We found several stable structures where the F- ion successfully bonds into the EC. But some of them are unachievable by a single transition state, based on the invading F- ion that "jumps" over the EC and ends at the other side. There are only 4 achievable final structures, one for each case, shown in **Figure 2**. From these structures as final states and their corresponding initial states, we calculate the transition states that are indicated by imaginary frequency existence. Accordingly, we proposed the decomposition mechanism of each case as below.

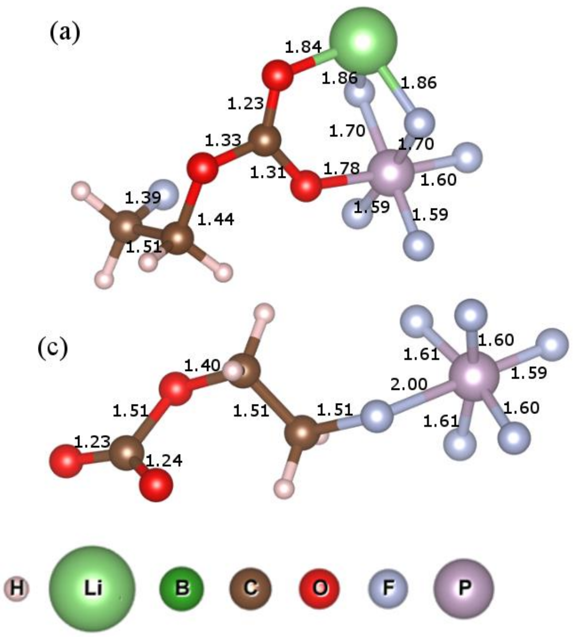

Figure 2. Final states of EC + (a)$LiPF_6$, (b)$LiBF_4$, (c)PF_6^-, and (d)BF_4^- with bond lengths (Å).

3.1.1. Decomposition mechanism of EC + $LiPF_6$

The EC molecule decomposes by rotation of EC molecule, thus approaching the ethylene side to the salt anion while breaking the O_E-C_E bond. After the ring opens, the nearest F atom attacks and bonds to the ethyl end. Then the carbonate end, attracted by the Li^+ ion, pulls into $LiPF_5^+$ whilst the fluorinated ethyl end rotates away from the carbonate end. **Figure 3** shows the decomposition mechanism of EC + $LiPF_6$.

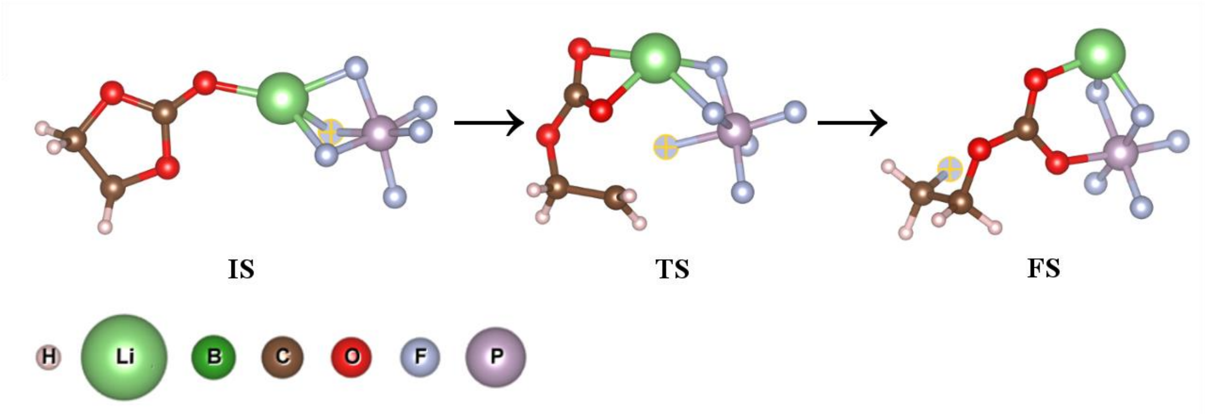

Figure 3. EC+$LiPF_6$ decomposition, the attacking F^- atom is highlighted.

3.1.2. Decomposition mechanism of EC + $LiBF_4$

Similar to the $LiPF_6$ case, the decomposition of EC + $LiBF_4$ occurs by approaching the ethylene side to the BF_4 while the ring opens. The nearest fluor atom bonds into the ethyl end and then rotates away with the carbonate end pulled back to the LiBF3. The decomposition mechanism is shown in **Figure 4**.

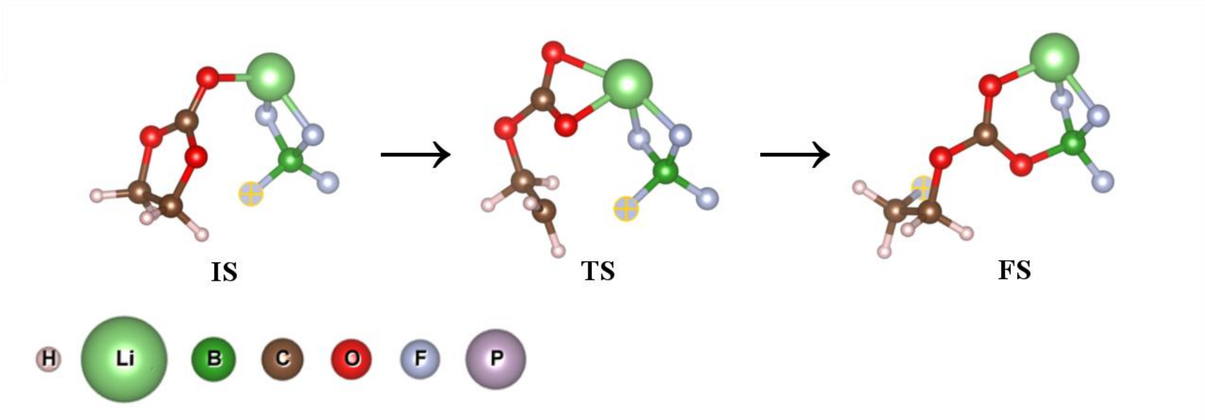

Figure 4. EC+$LiBF_4$ decomposition, the attacking F^- atom is highlighted.

3.1.3. Decomposition mechanism of EC + PF_6^-

In this case, since the ethylene side is already facing the PF_6^- anion, the ring opens directly with the ethyl end while rotates into the PF_6^- and bonds with the nearest F atom. The system stabilizes by the carbonate end rotates away, repulsed by the negative charge of PF_6^-. **Figure 5** shows this decomposition mechanism.

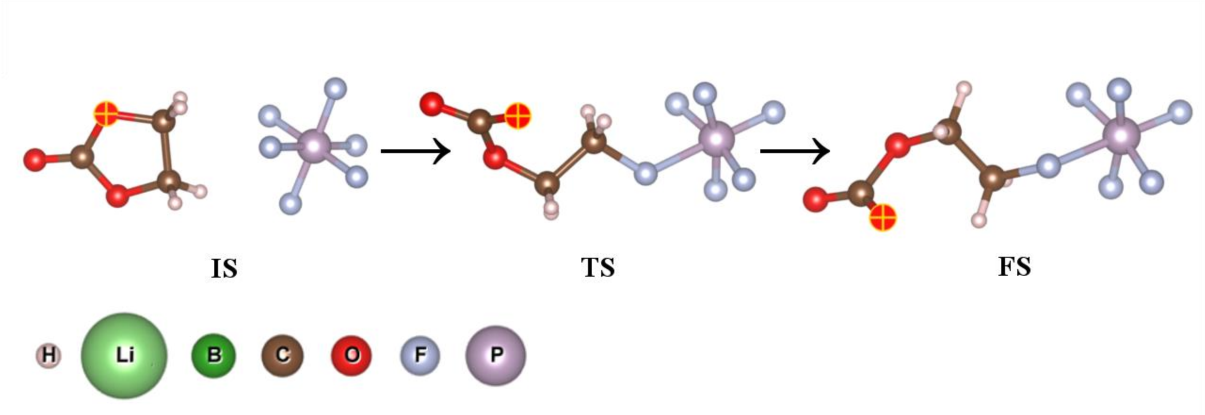

Figure 5. EC+PF_6^- decomposition, the O_E atom is highlighted to show carbonate rotation.

3.1.4. Decomposition mechanism of EC + BF_4^-

The same occurrence as the EC + PF_6^- case happened here. The already ethylene-end-facing-to-the-PF_6^- EC breaks its ring structure and its ethyl end catching the nearest F atom. The resulting 2-fluoroethyl carbonate structure relaxes to the configuration where the carbonate end is pushed to the furthest from the attached F atom. The decomposition mechanism is shown in **Figure 6**.

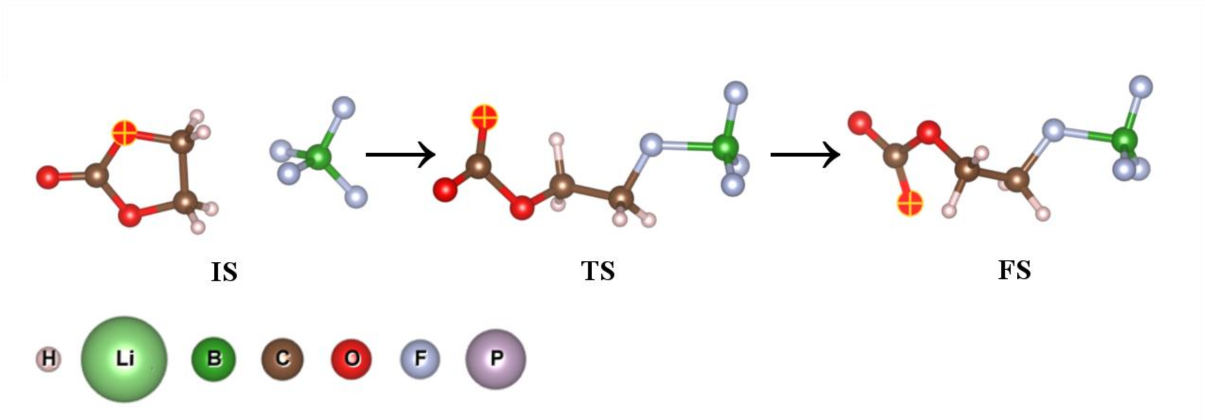

Figure 6. EC+BF_4^- decomposition, the O_E atom is highlighted to show carbonate rotation.

3.2. Energy barriers and enthalpy changes

The energy barriers and enthalpy changes for each decomposition mechanism's cases were determined from the system energies relative to the initial state energies. These results are shown in **Figure 7**. Another study has performed DFT calculations on EC+PF6$^-$ systems, but their results differ a lot from our results [14]. This dissimilarity comes from the different assumptions on the system, where Li et. al.'s work assumed an oxidized EC. In this study, we assumed a neutral charge system since any excess charges could move freely in liquid electrolyte systems.

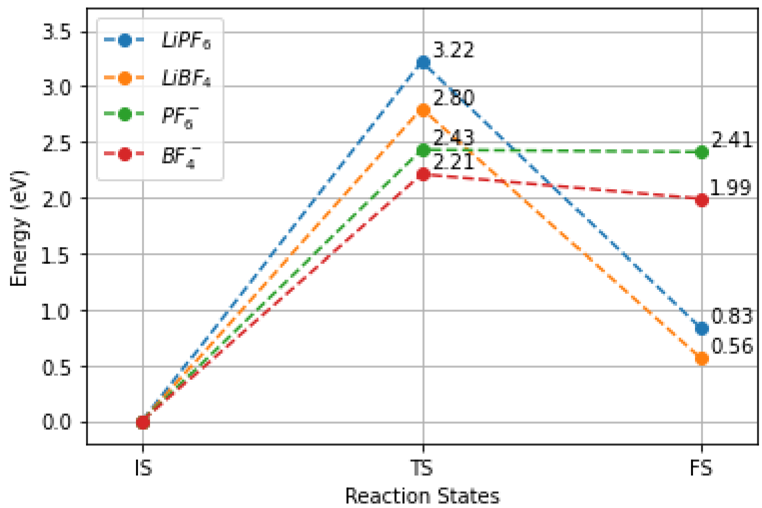

Figure 7. Reaction energies for all lithium salt cases.

3.3. Comparison of different salt anion

The main difference between PF_6^- and BF_4^- salt anion cases is the mechanism of bond breaking. Theoretically, the P-F bond is weaker than the B-F bond [18]. Contrarily, many studies have shown experimentally that $LiBF_4$ is more thermally stable than $LiPF_6$ [5–8]. Also, the BF_4^- anion is more table against hydrolysis than PF_6^- [7]. As a result, $LiPF_6$ is predicted to have lower stability (hence a lower energy barrier and smaller enthalpy change) than $LiBF_4$.

However, our calculation shows contrast results from the prediction above. In all cases, the transition states and final states of the $LiPF_6$ and PF_6^- cases have higher relative energy than the $LiBF_4$ and BF_4^- cases (**Figure 8)**. The energy barrier of $LiBF_4$ is 0.42 eV lower than $LiPF_6$, and in their delithiated case is 0.22 eV lower. The enthalpy change of $LiBF_4$ is also lower about 0.27 eV than

$LiPF_6$, and in their delithiated case is 0.42 eV lower. This indicates that EC with $LiBF_4$ and BF_4^- decompose easier and at a higher rate than with $LiPF_6$ and PF_6^-. We can conclude that the presence of BF_4^- anion destabilizes the EC more than PF_6^- anion, not caring whether lithium ions are present or not.

The conflicting results from our calculation and the experimental prediction could be caused by solvent effects. These effects were not included in our calculation since we use a vacuum condition.

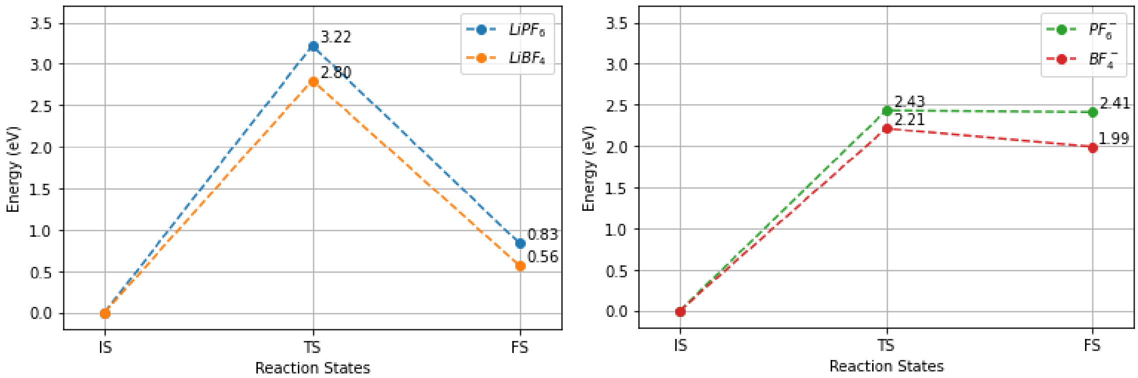

Figure 8. Comparison of $LiPF_6$ – $LiBF_4$ (left) and PF_6^- – BF_4^- (right) reaction energies.

3.4. *Effects of the lithium-ion presence*

The lithium-ion presence attracts the negative dipole charge on the carbonate side of EC thus making the ethylene side not facing directly to the PF_6^-. This initial configuration forces the EC to rotate first so that the ethylene side can catch the F atom. The additional rotation needed creates a higher energy barrier (about 0.79 eV) than in the delithiated case, where the EC could catch the F atom without rotating first. While the enthalpy change of the $LiPF_6$ case is significantly lower (about 1.58 eV) than its delithiated case. This may come from the strong dipole-dipole bonds from Li-O and P-O that create a more stable final structure than in the delithiated case which has fewer bonds in the final structure.

Similar effects are also seen in the BF_4^- case. The initial rotation of EC needed creates a higher energy requirement (about 0.79 eV), and the resulting bond of Li-O and B-O stabilizes the final structure more (a lower enthalpy change of 1.43 eV) than on its delithiated counterpart case. The changes of energy barrier and final state relative energy by Li^+ presence is shown in **Figure 9**.

In both cases, while the energy barrier increases and slows the decomposition rate, the decreasing enthalpy change suggests a more destabilized EC. Moreover, the energy change of transition and final states on the Li^+ absence case is relatively small. Therefore, the presence of Li^+ destabilized the EC, making it more likely to decompose.

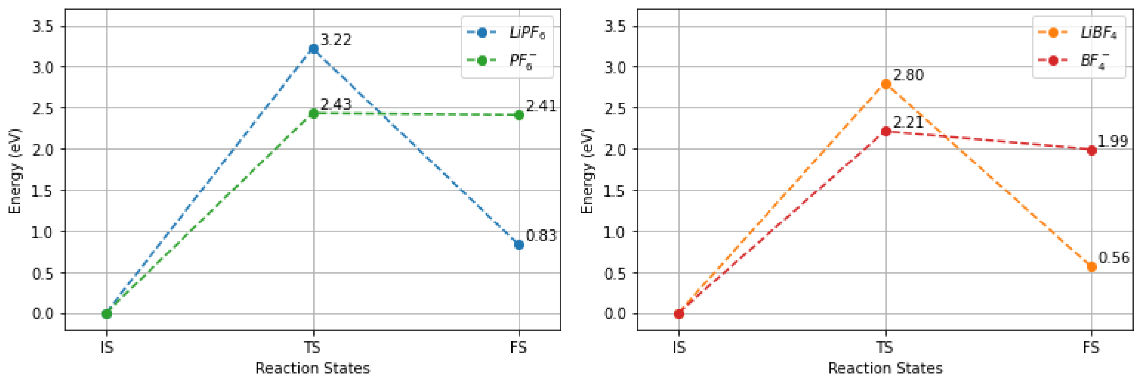

Figure 9. Comparison of $LiPF_6$ – PF_6^- (left) and $LiBF_4$ – BF_4^- (right) reaction energies.

4. Conclusion

DFT calculations on EC decomposition with $LiPF_6$ and $LiBF_4$ salt have given several insights into the mechanisms and stability of these systems. It is shown that Li^+ ion presence increases the energy barrier of decomposition while at the same time decrease the enthalpy change. $LiPF_6$ has a higher energy barrier of decomposition (0.79 eV difference) but lower enthalpy change (1.58 eV difference) than PF_6^-. Similarly, $LiBF_4$ has a higher energy barrier of decomposition (0.59 eV difference) but lower enthalpy change (1.43 eV difference) than BF_4^-. This suggests that while the Li^+ ion decreases the decomposition rate, its presence destabilizes the EC more strongly than on Li^+ absence.

However, our results showed that $LiBF_4$ has a lower energy barrier of decomposition and enthalpy change, respectively, about 0.42 eV and 0.27 eV than $LiPF_6$. In their delithiated case, BF_4^- also have a lower energy barrier of decomposition and enthalpy change, respectively, 0.22 eV and 0.42 eV than PF_6^-. This suggests that $LiBF_4$ and BF_4^- presence reduce the stability of EC more strongly than $LiPF_6$ and PF_6^-, a disagreement on experimental results which shows superior thermal stability of $LiBF_4$ than $LiPF_6$. This may be caused by solvent effects that are not calculated by our vacuum condition model.

Hence, we suggest that a more realistic mix solvent model to be included in future research. By this solvent effect, the DFT simulation results may be able to explain experimental results on EC and lithium salts decompositions.

Acknowledgement

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